

NEHA SHETHIA

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SUMMARY

Software Engineer with 2+ years building high-performance distributed systems for finance platforms. Reduced system latency by 40% through microservices optimization and automated CI/CD workflows. Expertise in C/Java backend development and cloud infrastructure with focus on scalability and reliability.

PROFESSIONAL EXPERIENCE

Software Engineer Intern, Amazon Web Services | Seattle, WA

May 2024 – Aug 2024

- **Backend Development:** Designed and shipped **REST** based **APIs** to expose low-level query control, empowering customers to self-manage execution plans and reducing customer-reported issues by **30%**.
- **Performance Optimization:** Architected a query plan enforcement feature in **AWS Aurora PostgreSQL** using **C**, enabling enterprise customers to lock execution strategies and avoid fallback to inefficient plans reducing query latency by **40%**
- **System Performance & Reliability:** Engineered a robust solution to improve plan stability and fault tolerance, reducing plan failures by more than **50%** and optimizing **query performance** for enterprise clients.
- **CI/CD Workflows:** Automated **unit**, **integration**, and **end-to-end** (E2E) tests using (JUnit) with **90%+** test coverage for database query flows, integrating with **AWS CodePipeline** to enable continuous delivery using continuous integration and continuous deployment.

Software Engineer, Tata Consultancy Services | Mumbai, India

June 2021 – July 2023

- **Distributed Systems:** Developed a high-availability **SaaS** process in **Linux** environment using **C** language for real-time trade processing **100,000+** transactions/sec, leveraging **Terraform** to reduce downtime for **250+** clients in volatile market conditions.
- **Disaster Recovery Automation:** Built **Bash**-based recovery workflows for **MySQL** systems, improving uptime by **62%** and accelerating resolution of critical production incidents to ensure business continuity during market hours.
- **Full-Stack Development:** Engineered backend microservices in **Java (Spring Boot)** with **Object-Oriented Design**, exposing **RESTful APIs** and integrating with a **React** frontend, accelerating data retrieval by 40% .
- **Agile Development & Delivery:** Managed **Agile** Software Development life cycle (**SDLC**) for financial software, leveraging **Git** workflows and stakeholder collaboration to reduce post-release issues by 40% and accelerate feature rollout, improving delivery reliability and customer satisfaction.
- **Deployment Automation:** Spearheaded **CI/CD** automation for microservices using **Jenkins** and **GitHub Actions**, deploying to **Kubernetes** and reducing release time by 55%.

TECHNICAL SKILLS

Programming Languages & Concepts: Java, C/C++, Python, JavaScript, TypeScript, HTML, CSS, Object-Oriented Programming (OOP), Design Patterns, SOLID Principles, Test-Driven Development (TDD)

Frameworks: Spring Boot, Spring MVC, Node.js, Express.js, React, Next.js, React Native, Figma

DevOps & Tools: Kubernetes, Docker, GitHub Actions, CI/CD, Elasticsearch, Redis

Cloud Technologies: AWS (EC2, S3, Elastic Load Balancer, VPC, RDS, Route 53, IAM, CloudWatch, DynamoDB, KMS, Auto Scaling)

Databases: MySQL, Postgres, NoSQL, MongoDB, GraphQL, Microsoft SQL Server, RDBMS, Redis, Oracle

Testing: Unit/Integration/E2E Testing, Cypress, Cucumber, GoDog, TDD/BDD

PROJECTS

ABU-Robocon - Embedded C, OpenCV

- Secured **National Rank 5** in the International ABU-Robocon 2019 robotics competition organized by IIT-Delhi and collaborated with cross-functional teams on mechanical and electronic aspects of the robot design.
- Implemented image processing for identifying colors of the arena; interfaced 10+ industrial sensors in **Embedded C** for smooth operation of autonomous and 4 legged-quadruped robots.

AWS Cloud Application - AWS, Terraform, Node.js, CI/CD, GitHub Actions, Packer

- Architected and deployed a cloud-based web application using **Node.js** on AWS, hosting RESTful APIs on **EC2** instances using Terraform for **Infrastructure as Code** (IaC).
- Achieved **99%** uptime and ensured robust scalability across multiple availability zones with seamless auto-scaling for workloads.
- Incorporated **AWS Lambda** functions for automated email alerts, **RDS** for database management, **S3** for user profile image storage and **AWS CloudWatch** for continuous monitoring enhancing application performance and reliability.

EDUCATION

Northeastern University | Boston, MA

Aug 2023 - May 2025

Master of Science, Computer Software Engineering | GPA - 3.7/4.0

Roles: Lead Teaching Assistant for Object-Oriented Programming course.

K.J Somaiya College of Engineering | Mumbai, India

Aug 2017 - May 2021

B. Tech, Electronics and Telecommunication Engineering